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MIL-DTL-15665J
6 April 1998
SUPERSEDING
MIL-I-15665H
29 April 1994
MIL-I-3575F
5 January 1989

DETAIL SPECIFICATION

INSIGNIA, BRANCH OF SERVICE, HAT, GARRISON CAP AND SERVICE CAP, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers insignia used for branch of service, hat, garrison cap and service cap used by personnel of the Armed Forces (see 6.1).

1.2 Classification. The insignia covered by this specification will be as specified on the applicable military specification sheet (See 2.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in preparing this document should be addressed to: Director, The Institute of Heraldry, US Army, 9325 Gunston Road, Room S112, Fort Belvoir, VA 22060-5579, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

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specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-F-495 - Finish, Chemical, Black for Copper Base Alloy

FEDERAL STANDARDS

FED-STD-595 - Colors Used in Government Procurement

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 42, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DoDISS - Department of Defense Index of Specifications and Standards

(Copies of the DoDISS are available on a yearly subscription basis either from the Government Printing Office or the DoDSSP Subscription Services, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

SPECIFICATION SHEETS

(See Supplement 1 for list of specification sheets)

DRAWINGS

THE INSTITUTE OF HERALDRY

4-4-89 - Insignia of Branch, Enlisted Personnel, Assembly and Details

B-1 (Series) - Insignia, Cap

B-2 (Series) - Insignia, Branch of Service

B-13-12 - Attaching Devices, Heraldic, Prong and Clutch Type

STANDARD COLOR CHIP SET

THE INSTITUTE OF HERALDRY, U.S. ARMY STANDARD HARD ENAMEL COLOR CHIP SET

THE INSTITUTE OF HERALDRY, U.S. ARMY STANDARD METAL FINISH CHIP SET

(The standard color chip(s) for color or finish may be obtained from the procuring activity for Government procurements.)

(Copies of specifications, standards, drawings and specification sheets required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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(Figure 1 and Figure 2 are miniature copies of Institute of Heraldry drawings and are for information only.)

2.3 Non-Government publications. The following documents form a part of this document to the extend specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are issues of the documents cited in the solicitation (see 6.2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-B-86 - Zinc-Alloy Die Casting
ASTM-B-487 - Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section, Measurement of
ASTM-D-2240 - Rubber Property - Durometer Hardness

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheets, the latter shall govern. The design of the insignia shown on the specification sheet are for illustrative purposes only.

3.2 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Recycled, recovered or environmentally preferable materials. Recycled, recovered, or environmental preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials. Materials shall as specified herein. Recovered material shall be used to the maximum extent possible.

3.4.1 Copper base alloy. The copper base alloy shall be roll polished, free from pits, scale (including red oxide), dents, nicks, cracks, scratches, segregation and foreign inclusions that will not be removed in later processing. When tested as specified in 4.5.1, the

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chemical composition of the copper content of the copper base alloy shall be as specified in Table I.

TABLE I. Copper content of copper base alloys.

ALLOY	COPPER
Yellow Brass	64.0-68.5
Red Brass	84.0-86.0
Gilding Metal	94.0-96.0
Free Cutting Brass	60.0-63.0
Bronze	89.0-91.0
Nickel Silver	63.0-65.5 1/, 2/
Low Brass	78.5-81.5

1/ Copper content for the nickel silver pin only may be 53.6 to 56.5%.
2/ When nickel silver is used as a base material, the nickel content shall be not less than 18%.

3.4.2 Gold plating.

3.4.2.1 24 karat gold for plating. Gold for plating shall be 24 karat.

3.4.2.2. 22 karat gold for plating. When specified on the applicable specification sheet, gold for manufacturing insignia with nickel underplating shall be 22 karat.

3.4.3 Zinc aluminum alloy. The alloy used for die casting shall conform to alloy AG 40a or AC 41a of ASTM-B-86.

3.4.4. Nickel for plating. Nickel for plating shall be the nickel plating normally used in commercial practice.

3.4.5 Silver for plating. Silver for plating shall be not less than 99.5 percent pure silver.

3.4.6 Rhodium Plating. Rhodium for plating shall be the rhodium plating normally used in commercial practice.

3.4.7 Solder.

3.4.7.1 Hard Solder. Hard solder shall be a silver alloy solder having a melting point of not less than 1075 degrees Fahrenheit (561.7 degrees Celsius).

3.4.7.2 Soft Solder. Soft solder shall be a lead-tin alloy having a melting point of not less than 375 degrees Fahrenheit (190.58 degrees Celsius).

3.4.8 Enamel. All enamel shall be lead free.

3.4.8.1 Hard enamel. Hard enamel shall be a glass, vitreous type enamel fused with metallic oxides to produce the required color, opacity or translucence and shade.

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3.4.8.2 Epoxy resin (stoning epoxy). Epoxy resin shall be a stoning epoxy, have a shore D hardness of 85 and incorporate an ultraviolet inhibitor.

3.4.8.3 Baking enamel. The enamel for subdued insignia shall be a flat synthetic, baking enamel pigmented to the shade of black matching the standards established in FED-STD-595 for Color Chip number 37038.

3.4.8.4 Soft enamel. Soft enamel shall be a baking enamel pigmented to produce the required color or shade.

3.4.9 Lacquer. Lacquer shall be a pale clear synthetic lacquer. The use of a pigmented lacquer shall not be permitted.

3.5 Design. The embossed design of the insignia shall be an exact duplicate of the design on the Government loaned hubs (see 3.8), from which the contractor's working dies shall be extracted. The contractor's working die shall be tooled and polished to remove any dents, nicks, scratches or other imperfections.

3.6 Construction. Army enlisted branch of service insignia shall be constructed in accordance with details and dimensions shown on Drawing 4-4-89 (Figure 1).

3.6.1 Stamping, trimming and piercing. The stamping shall have a well defined die struck edge. The insignia shall be trimmed and, when applicable, pierced to the die struck edge. All edges shall be clean, smooth and free from burrs, drag, step and rough edges. The stamping, piercing and trimming operations shall not damage or distort the design or alter the shape of the item.

3.6.2 Die casting. When die cast method is applicable, the insignia shall be free from porosity, blow holes or other casting defects. Before finishing, all flash lines shall be removed and the edges of the insignia shall be smooth. Insignia that is die cast from zinc aluminum specified in 3.4.3 shall be copper plated. When tested as specified in 4.5.2.2, the plating shall be a minimum of 0.0003 inch thick (0.007 cm).

3.6.3 Soldering

3.6.3.1 Hard soldering and electronic fusion. Unless otherwise specified on the applicable specification sheet, all soldering shall be accomplished using hard solder specified in 3.4.7.1 or by electronic fusion. Joints shall be clean, smooth, strong and free from burned or reduced areas. There shall be no excess solder and all excess flux shall be removed. When hard solder is used, the soldered part and prong shall not separate at the joint when tested as specified in 4.5.4.1. When fused joints are used, the prong shall not separate at the joint when tested as specified in 4.5.5.

3.6.3.2 Soft Soldering. When soft solder is specified, soft solder specified in 3.4.7.2 shall be used. When solder is specified for joining a superimposed device to the insignia, the soldering shall be accomplished using soft solder specified in 3.4.7.2 and complete contact shall be made between the soldered parts. When tested as specified in 4.5.4.2, the soldered device shall not separate.

3.6.4 Forcers. When forcers are permitted to bring up the design, the back of the insignia shall have no sharp or rough edges. The impression of the forcer shall be symmetrical with the embossed design of

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the insignia. Sufficient metal shall be left near any trimmed area to assure clean smooth trimming without distortion. Unless otherwise specified on the applicable specification sheet, mock forcers shall not be permitted. When mock forcers are permitted, the forcer shall be used only on the areas specified.

3.6.5 Attaching device. Attaching devices shall be as specified on the applicable specification sheet except for Army enlisted branch of service and collar insignia which shall have a back plate with prong and clutch type attaching device as specified on Drawing 4-4-89 (Figure 1). The back plate shall be fabricated of any type of 1/4 hard brass. As an alternative, the back plate and prongs may be die cast as specified in 3.6.2; however, the back plate shall be brass plated after copper plating. When clutches are tested in accordance with 4.5.9.1, it shall not be possible to remove the clutch from the prong without first depressing the release wings.

3.6.5.1 Prong (registering). Prongs shall be made from nickel silver specified in 3.4.1. The prongs shall be of the dimensions shown on the applicable specification sheet. They shall be positioned and attached as required by the applicable specification sheet. Registering prongs shall be an integral part of the insignia when die cast zinc aluminum is used (see 3.4.3).

3.6.5.2 Prong (integral). When prongs are die cast as integral part of the insignia and tested in accordance with 4.5.2.1, no evidence of breaking or cracking shall occur.

3.6.5.3 Screw post type and thumb nut. Screw post and thumb nut attaching devices shall be made from free cutting brass specified in 3.4.1 and shall conform to the details and dimensions shown on the applicable specification sheet. The screw post shall be hard soldered or electronically fused to the back of the insignia in the location shown. When tested as specified in 4.5.9.2, thumb nuts shall not bind on the post's threads. Screw post shall be an integral part of insignia when die cast zinc aluminum is used (see 3.4.3).

3.6.5.4 Prong and clutch type. Unless otherwise specified, prongs as shown on Drawing B-13-12 (Figure 2) shall be made from nickel silver specified in 3.4.1, shall be 5/16 inch ± 1/64 inch (0.79 cm ± 0.04 cm) long and shall be hard soldered or electronically fused in the locations specified on the applicable specification sheet. Clutches as shown on Drawing B-13-12 (Figure 2) shall be made from any type brass. When tested as specified in 4.5.9.1, it shall not be possible to remove the clutch from the prong without first depressing the release wings. When prongs are die cast as an integral part of the insignia and tested in accordance with 4.5.2.1, no evidence of breaking or cracking shall occur.

3.6.5.5 Back plate with prong and clutch. The back plate with prong and clutch type attaching device shall conform to the dimensions shown on Drawing 4-4-89 (Figure 1). The back plate shall be fabricated from any type 1/4 hard brass. The prongs shall be 5/16 inch ± 1/64 inch (0.79 cm ± 0.04 cm) long, be made from hard drawn brass and be driven, swaged, hard soldered or electronically fused in the locations specified in Drawing 4-4-89 (Figure 1). Clutches as shown on Drawing B-13-12 (Figure 2) shall be made from any type brass. As an alternative, the back plate and prongs may be die cast as specified in 3.6.2; however, the back plate shall be brass plated after copper plating. When clutches are tested in accordance with 4.5.9.1, it shall not be possible to remove the clutch from the prong without first depressing the release wings.

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3.7 Finish. Unless otherwise specified, the front and edges (surfaces visible from the front) shall be finished as specified on the applicable specification sheet and shall match the TIOH standard enamel color, epoxy color or metal finish chips as required (see 2.2.2). Where plating or oxidizing is specified, the entire item (front, back and edges), except attaching devices, shall be plated or oxidized. Unless otherwise specified, the backs of all oxidized items shall be clean and uniformly oxidized. Army enlisted branch of service insignia and collar insignia may be gold plated for commercial sale. In addition to the finish specified, all insignia, except Marine Corps black enameled insignia and gold plated insignia, shall be thoroughly coated (front, back and edges) with clear lacquer specified in 3.4.9. Army enlisted branch of service and collar insignia need only be lacquered on the front surface and edges. At the option of the manufacturer, any gold plated insignia may be lacquered. Marine Corps black colored insignia shall be given a black chemical finish (front, back and edges) conforming to MIL-F-495. The face and edges of the insignia shall then be coated with flat dull black baking enamel specified in 3.4.8.3.

3.7.1 Plating. All plating shall be by electroplating methods. The plating shall be continuous and unbroken over the entire plated surface. There shall be no cut-through, shade, peeled, or blistered plating. Plating of attaching devices will not be required.

3.7.1.1 24 karat gold plating. The gold plating shall be accomplished using gold specified in 3.4.2.1. The use of nickel as an undercoating with 24 karat gold shall not be permitted. When tested as specified in 4.5.3, no visible chemical reaction such as evolution of gases shall appear.

3.7.1.2 22 karat gold plating. The gold plating shall be accomplished using gold specified in 3.4.2.2. The use of nickel as an undercoating shall be permitted. When tested as specified in 4.5.3, no visible chemical reaction such as evolution of gases shall appear.

3.7.1.3 Nickel plating. Nickel for plating shall be the nickel specified in 3.4.4 and shall be not less than 0.0003 inch (0.0008 cm) thick. Testing shall be standard commercial. In case of a dispute, testing shall be in accordance with ASTM-B-487.

3.7.1.4 Nickel underplating. When nickel underplating is specified, nickel specified in 3.4.4 shall be used and shall be not less than 0.00025 inch (0.000635 cm) thick. Testing shall be standard commercial. In case of a dispute, testing shall be in accordance with ASTM-B-487.

3.7.1.5 Silver plating. Silver plating shall be accomplished using the silver specified in 3.4.5. After finishing, the silver plating shall be a minimum of 0.0003 inch (0.0008 cm) thick. Testing shall be standard commercial. Items made of brass shall be silver thickness tested on the obverse side. In case of a dispute, testing shall be conducted in accordance with ASTM-B-487.

3.7.1.6 Rhodium Plating. When rhodium plating is specified, rhodium specified in 3.4.6 shall be used. When tested as specified in 4.5.3, no chemical reaction shall occur. When rhodium plating is required over brass, nickel underplating specified in 3.4.4 shall be required.

3.7.2 Enameling. Unless otherwise specified on the applicable specification sheet, all enameling shall be accomplished using hard enamel

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specified in 3.4.8.1 or epoxy specified in 3.4.8.2. The enamel or epoxy shall be applied so as to be uniform in color, free from bubbles, foreign inclusions, cracking, crazing, or other defects which might affect appearance. There shall be no overrunning of enamel or epoxy.

3.7.2.1 Colors. Enamel or epoxy colors shall match the shades of the Institute of Heraldry color chips specified on the applicable specification sheet (see 2.2.2).

3.7.2.2 Hard enameling. Hard enamel specified in 3.4.8.1 shall be charged, fired and stoned level with the dikes. No design elements shall be removed during the stoning process. There shall be no noticeable burn spots as a result of the firing. The hard enamel shall then be lapped, felted, or wheel polished to produce a glass like finish. The finish shall be adherent and free from bubbles, pits, foreign inclusions, cracking, crazing, burned edges or darkened enamel.

3.7.2.3 Epoxy Resin. Epoxy resin specified in 3.4.8.2 shall be applied and stoned level with the dikes. No design elements shall be removed during the stoning process. The epoxy resin shall then be lapped, felted, or wheel polished to produce a glass like finish. When tested as specified in 4.5.6, the epoxy shall have a shore D hardness of not less than 85.

3.7.2.4 Baking enamel. The face and edges of subdued insignia shall be coated with enamel specified in 3.4.8.3. Prior to enameling, the entire insignia shall be given a black chemical finish conforming to MIL-F-495. When tested as specified in 4.5.8, the baking enamel shall not flake or chip.

3.7.3 Cutting down. When necessary to remove nicks, scratches, pin holes or other blemishes, the insignia shall not be cut down to the extent that any details of the design are obliterated or reduced.

3.7.4 Lacquering. All insignia, except Marine Corps black enameled insignia, gold plated insignia and nickel plated insignia, shall be lacquered using lacquer specified in 3.4.9 as applicable (see 3.7 and appropriate specification sheet). The dry lacquer film shall be continuous, level, adherent and free from lint, dust or other foreign matter. When tested as specified in 4.5.7, the tissue paper shall not adhere to the clear lacquered surface.

3.8 Government loaned property. Hubs will be loaned by the Government and shall be used to make the contractor's working dies necessary for one contract or order (see 3.5 and 6.4).

3.9 Marking for identification. Except for the Air Force service cap insignia, Air Force branch of service insignia, Marine Corps branch of service insignia and female hat insignia, the contractor shall stamp his trade mark or other identifying mark and the letters "GI" (GI for government procurement only) legibly and inconspicuously on the back of each item. When practical, the items will be stamped in the forced areas. Marking shall be placed on the attaching bar for Army branch of service insignia and on the washer or thumb nut for the Air Force service cap insignia, Marine Corps branch of service insignia, and Marine Corps service cap insignia. Female hat insignia and Air Force branch of service insignia need not be marked.

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3.10 Marking for cleaning. Marking for cleaning of gold plated items shall be as follows: "To clean, rub gently with a soft, dry cloth. Do not clean with a chemically treated cloth or abrasive".

3.11 Workmanship. The finished item shall be clean, well made, and shall conform to the acceptable quality levels established by this specification.

3.12 Unit packing for insignia, garrison cap, Marine Corps. One Marine Corps garrison cap insignia (left hand only) shall be mounted on a white sulfite board card, 2 inches (2.54 cm) square and not less than 0.011 inch (0.028 cm) thick. Each card shall then be packaged in a snug-fitting transparent polyethylene bag. The polyethylene film shall have a minimum thickness of 0.002 inch (0.005 cm). All seams and closures shall be heat sealed. Prior to or during the heat sealing operation, excess air within the bag shall be expelled.

3.13 Insignia, service cap, Army; insignia, branch of service, insignia, service cap, Air Force; insignia, branch of service, collar and insignia, service cap, Marine Corps; and insignia, hat, Female, Army. One Army, Air Force or Marine Corps service cap insignia (one left hand and one right hand) or two Air Force branch of service insignia shall be mounted on a white sulfite board 2-1/2 inches (6.35 cm) square and not less than 0.011 inch (0.028 cm) thick. Each card shall then be packaged in a snug-fitting transparent polyethylene bag. The polyethylene film shall be a minimum of 0.002 inch (0.005 cm) thick. All seams and closures shall be heat sealed. Prior to or during the heat sealing operation, excess air within the bag shall be expelled.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in 4.4 and 4.5.

4.3 First article inspection. Inspection and testing of the first article (see 3.2) shall be made of a completely finished item for all provisions of this specification applicable to the end product examination and tests.

4.4 Conformance inspection. Inspection shall be in accordance with the provisions set forth herein, except where otherwise indicated (see 6.3).

4.4.1 Conformance inspection sampling.

4.4.1.1 Certification of compliance. Materials listed below may be accepted on the basis of the contractor's certificate of compliance for requirements specified in applicable paragraphs of this specification.

Table II. Contractor's certificate of compliance

COMPONENTS	CHARACTERISTICS	REQUIREMENT

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		PARAGRAPH
Copper base alloy	Material identification	3.4.1
Zinc aluminum alloy	Material identification	3.4.3
24 karat gold plating	Material identification and karat	3.4.2.1
22 karat gold plating	Material identification and karat	3.4.2.2
Nickel for plating	Material identification	3.4.4

Table II. Contractor's certificate of compliance (Continued)

Silver for plating	Material identification and fineness	3.4.5
Rhodium plating	Material identification	3.4.6
Hard solder	Material identification	3.4.7.1
Soft solder	Material identification	3.4.7.2
Hard enamel	Material identification	3.4.8.1
Soft enamel	Material identification	3.4.8.4
Epoxy resin	Material identification	3.4.8.2
Baking enamel	Material identification	3.4.8.3
Lacquer	Material identification	3.4.9
Brass for prongs (when applicable)	Material identification	3.6.5.5
Nickel silver for prong (when applicable)	Material identification	3.6.5.1 and 3.5.6.4
Brass for screw post & thumb nut	Material Identification	3.6.5.3
Brass for washer (Marine Corps only)	Material identification	As specified on applicable specification sheet
Back Plate (Army branch of service insignia)	Material identification and temper	3.6.5.5

4.4.1.2 In-process inspection. In process inspection shall be made at any point or during any phase of the manufacturing process to determine whether operation or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance. In-process inspection shall be conducted to see that accomplished of the following is in accordance with the specification requirements.

REQUIREMENT, OPERATION OR ASSEMBLY	CHARACTERISTICS	REQUIREMENT PARAGRAPH
Black chemical finish	Used as a base	3.7.2.4
Nickel plating	Not used as an underplating	3.7.1.1

4.4.2 Inspection of the end item.

4.4.2.1 Visual examination of insignia. Visual examination of insignia for defects in finish, design, material, construction, workmanship and marking shall be made using Table II. Examination shall be made at a distance of approximately 16 to 18 inches with illumination equal to average daylight and arranged so as to avoid as much reflected light as possible. Defects designated by an asterisk (*) shall be classified a major when seriously affecting appearance or serviceability and minor when affecting appearance or serviceability but not seriously.

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TABLE II. Defects.

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	(*)	MINOR
Color & Finish	Poor match to the TIOH Standard Metal Finish Chip	X	–	–
	Not required finish	–	*	–
	Discoloration, such as spot, stain or speck	–	*	–
	Not clean	–	–	*
	Paired insignia not matching in color, shade and finish (when applicable	–	*	–
	Any solder spatter clearly noticeable	–	*	–
	Background not finished as specified	X	–	–
	The oxidized finish, when required, has not been sufficiently relieved, or has been excessively relieved	–	*	–
	Not highlighted to the same extent as the TIOH Metal Finish Chips	–	*	–
	Background or recessed areas are not finished as specified	X	–	–
	Surface contains pits, scale, nick, scratch, crack, machine mark, pin hole, segregation, rupture, foreign matter or other blemish	–	*	–
	Discoloration, spot, stain or speck affecting appearance	–	*	–
	Buff drag, blushed, or cloudy finish	–	*	–
Hard enamel or epoxy	Not enameled (when required)	X	–	–
	Wrong color or shade	X	–	–
	Cracked, chipped or crazed	–	*	–
	Not stoned smooth, i.e., is coarse, uneven or has drag marks	–	*	–
	Not level with dikes, i.e., contains high or low spots	–	*	–
	Hole or blister	–	*	–
	Laps, overruns or skips	–	*	–
	Not clean, i.e., blemished such as spot, stain, spatter or imbedded foreign matter	–	*	–
Soft enamel	Not enameled or area of no film	X	–	–
	Area of thin film	–	–	X
	Enamel not flat and dull black	X	–	–
	Not adherent, i.e., blistered, flaking or peeling	–	*	–
	Foreign matter imbedded	–	*	–
	Discoloration clearly noticeable	–	*	–
	Coating not continuous and level, i.e., runs, drips, or drops	–	*	–

TABLE II. Defects (Continued).

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	(*)	MINOR
Design	Construction or design details			

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	obscured, altered or not conforming to government hub	X	–	–
	Not dapped (when required)	X	–	–
	Any significant detail not clear, missing or obliterated	X	–	–
	Any significant detail marred or reduced	–	*	–
	Any warp, twist or distortion, irregular surface contour or outline	–	*	–
	Any detail struck over resulting in a double impression	–	*	–
	Any operation not in accordance with the specified requirements	–	*	–
	Any significant detail not clear, marred, missing, obliterated or reduced	X	–	–
	Any area not pierced (when required)	–	*	–
Plating	Not plated (when required)	X		
	Not type specified	X		
	Plating is not smooth, continuous or adherent	X	–	–
	Any plated area cut through, porous, pitted, crystallized flaking, peeling or blistered	X	–	–
	Foreign matter imbedded	–	*	–
Hard enamel and epoxy	Not enameled (when required)	X	–	–
	Wrong color or shade	X	–	–
	Cracked, chipped or crazed	–	*	–
	Not stoned smooth, i.e., is coarse, uneven or has drag marks	–	*	–
Workmanship and construction	Surface or edge not clean, not smooth or free construction from burrs, roughness, drag, step or tool marks .	–	*	–
	Metal marks on exposed surface such as nick, dent, dig, gouge or scratch	–	*	–
	Die-struck edge, rim or outline is not defined when applicable	X	–	–
	Not trimmed to the die-struck edge, when applicable	–	*	–
	Pierced edge not clean and smooth, i.e., any burr, drag, cutter, or file mark	–	*	–
	Design or shape is damaged, distorted or altered by piercing operation	X	–	–

TABLE II. Defects (Continued).

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	(*)	MINOR
	Forcer used when not specified or insignia forced in other than specified location	–	*	–
	Impression of the forcer (when applicable) is not symmetrical with the design on the front	–	–	X
	Metal torn or broken out by use of			

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	forcer	–	–	X
	Any component missing, bent, twisted, broken or malformed	–	*	–
	Solder does not completely unite parts, i.e., any perceptible opening between component parts	–	*	–
	Area burned or reduced in soldering .	–	*	–
	Soldered joint not clean or smooth ..	–	–	X
	Excess solder or excess flux not removed	–	–	X
	Operations are poorly accomplished ..	–	*	–
	Letters distorted	–	*	–
	Back not flat or smooth when required			X
Attaching device	Defective, i.e., any part damaged or malformed which affects use or which may be expected to affect use	X	–	–
	Deformed, i.e., any part misshapen, damaged, bent, but not affecting use	–	–	X
	Not type specified	X	–	–
	Not positioned as specified	X	–	–
	Not attached in specified manner, i.e., not soldered or swaged as required.	X	–	–
	Metal strip not clinched around shank prong (when required)	X	–	–
	Clutch fastener will not engage or is loose fit on prong	X	–	–
	Clutch pierced through by prong	–	*	–
	Swaged prong can be removed by hand .	X	–	–
	Both swaged prongs are loose and wobble	–	–	X
	Threads on post or nut are stripped, crossed, damaged or not specified size or type	X	–	–
	Any component omitted, misplaced, operation omitted or not properly performed	–	*	–
Identification Marking	Missing, incorrect, illegible, misspelled, not accomplished as specified, or not placed as required	–	–	X

TABLE II. Defects (Continued).

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	(*)	MINOR
Lacquer	Not lacquered when required	X	–	–
	Areas of no lacquer	–	*	–
	Lacquer forms noticeable runs or sharp coarse particles	–	*	–
	Foreign matter embedded in finish, i.e., lint, dust, etc., clearly noticeable	–	*	–
	Hazy, rainbow effect, cloudy, or powdering	–	*	–
	Not smooth, continuous or adherent, i.e., flaking, blistering, peeling, rippling or has run, sag or drops ...	–	*	–
	Not set to touch, i.e., tacky when			

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	pressure is applied to coating	-	-	X
Quality of unplated metal	Surface pitted, porous, crystalline, spotted or opened grained, or otherwise impaired	X	-	-

4.4.2.1.2 Examination of decorations for defects in dimensions. Any dimension which is not within the specified tolerance shall be classified as a defect.

4.4.2.2 End item testing. Testing of the completely fabricated insignia shall be performed in accordance with Table III for the characteristics shown therein. The sample unit shall be one completely fabricated insignia. The results shall be reported as pass or fail for the examination.

TABLE III. Testing of end item.

CHARACTERISTIC	REQUIREMENT PARAGRAPH	TEST METHOD	# DETERM PER SAMPLE /UNIT	RESULTS REPORTED AS
Test for copper content	3.4.1	4.5.1	1	Pass or Fail
Test for fused joints:	3.6.3	4.5.5	1	Pass or Fail
Test for soldering:				
Hard soldered joints	3.6.3.1	4.5.4.1	1	Pass or Fail
Soft soldered joints	3.6.3.2	4.5.4.1	1	Pass or Fail
Test for attaching devices:				
Integral prong	3.6.5.2 1/	4.5.2.1	1	Pass or Fail
Prong and clutch	3.6.5.4	4.5.9.1	1	Pass or Fail
Screw post & thumb nuts	3.6.5.3	4.5.9.2	1	Pass or Fail
Test for fused joints:				
Prongs	3.6.3	4.5.5	1	Pass or Fail
Tests for enamels:				
Epoxy resin	3.7.2.3	4.5.6	1	Pass or Fail

TABLE III. Testing of end item (Continued).

Baking enamel	3.7.2.4	4.5.8	1	Pass or Fail
Tests For Plating:				
Nickel plating	3.7.1.3	STD comm'l	1	Pass or Fail
Nickel plating Thickness	3.7.1.3	STD comm'l	1	Pass or Fail
Nickel underplating	3.7.1.4	STD comm'l	1	Pass or Fail
Silver plating Thickness	3.7.1.5	STD comm'l	1	Pass or Fail
Acid Test:				
Gold plating, 24 Karat	3.7.1.1	4.5.3	1	Pass or Fail
Gold plating, 22 Karat	3.7.1.2	4.5.3	1	Pass or Fail
Rhodium	3.7.1.6	4.5.3	1	Pass or Fail
Lacquer tackiness test	3.7.4	4.5.7	1	Pass or Fail

1/ Insignia are to be tested per sample unit as two prongs per insignia.

4.5 Tests.

4.5.1 Copper content test for copper base alloy. Copper content shall be determined by standard commercial testing. Results shall be

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evaluated to determine compliance with the requirements specified in 3.4.1.

4.5.2 Test of Die Casting.

4.5.2.1 Test for integral prongs. Insignia to be tested shall be placed in a jig which allows no movement of the insignia and position prongs at a 30 degree angle above the horizontal. A vertical 10 pound load shall be applied to the top third of each prong. The prongs shall be examined to determine compliance with 3.6.5.2.

4.5.2.2 Direct micro thickness test for copper plating. Copper plating thickness shall be determined shall be determined by standard commercial testing. Results shall be evaluated to determine compliance with 3.6.2.

4.5.3 Acid test for gold and rhodium plating. If the test insignia has been lacquered, the lacquer shall be completely removed prior to testing with an organic solvent. The test acid shall be applied as follows: Place a drop of acid of not less than 1/16 inch (0.16 cm) in diameter on three (3) different spots on the plated surface allowing the drops to remain for not less than one (1) minute during which time the surface of the item shall be inspected to determine compliance with the requirements in 3.7.1.1 and 3.7.1.2. No visible chemical reactions shall occur. Acid drops to be placed on flat surfaces on the obverse or reverse face when possible. A minimum of two spots must withstand the acid test. The test acid shall be applied at room temperature (60 to 80 degrees Fahrenheit/15.6 to 26.7 degrees Celsius) and shall consist of a solution containing 50 percent by volume of chemically pure nitric acid (specific gravity 1.42) and an equal volume of distilled water.

4.5.4 Test for soldered joints.

4.5.4.1 Hard solder. Hard soldered items shall be placed in an oven maintained at 1075 degrees Fahrenheit $\pm$ 5 degrees Fahrenheit (561.7 degrees Celsius $\pm$ 2.78 degrees Celsius) for 15 minutes. While at this temperature the item shall be lifted by the attaching device and shall be inspected to determine compliance with the requirements specified in 3.6.3.1.

4.5.4.2 Soft solder. Soft soldered items shall be placed in an oven maintained at 365 degree Fahrenheit $\pm$ 5 degrees Fahrenheit (184.98 degrees Celsius $\pm$ 2.78 degrees Celsius) for 15 minutes. While at this temperature the item shall be lifted by the superimposed design and shall be inspected to determine compliance with the requirements specified in 3.6.3.2.

4.5.5 Test for electronically fused joints. Item to be tested shall be anchored on a horizontal surface. All prongs on each end item shall be grasped at least 1/3 its length above the fused joint and bent through an angle of 90 degrees (45 degrees to each side of the vertical) until the prong breaks. The fused joint shall then be examined to determine compliance with 3.6.3.1. A bending tool in the form of a 45 degree template may be used for this test provided the prong is gripped at least 1/3 its length above the fused joint.

4.5.6 Test for shore hardness of epoxy resin. Shore hardness shall be determined in accordance with ASTM-D-2240. Results shall be evaluated to determine compliance with the requirements of 3.7.2.3.

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4.5.7 Test for lacquer. At room temperature (60 to 80 degrees Fahrenheit/15.6 to 26.7 degrees Celsius), press a piece of tissue paper against the lacquered surface for 15 seconds, using any pressure capable of being exerted between thumb and two fingers, after which the pressure shall be released and the item inspected to determine compliance with 3.7.4.

4.5.8 Test for baking enamel. Draw a knife edge across the face side of the specimen making four parallel scores 1/16 inch (0.16 cm) apart. Repeat this process drawing the knife edge this time perpendicular to the previously scored lines, making small blocks in the finish. The finish shall then be examined to determine compliance with 3.7.2.4.

4.5.9 Test for attaching device.

4.5.9.1 Test for prong and clutch type. The clutches shall be removed and replaced ten times from the prong using the clutch release wings. An attempt shall then be made to remove the clutch by hand without first depressing the release wings. An inspection shall be made at this time of the clutch and prong to determine compliance with the requirements specified in 3.6.5, 3.6.5.4 or 3.6.5.5.

4.5.9.2 Test for screw post type and thumb nut. The thumb nut shall be assembled on the screw post, during which the operation of screw threads shall be observed to determine compliance with the requirements specified in 3.6.5.3.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Insignia covered by this specification are intended to be worn by personnel of the Armed Forces to indicate the wearer is a member of a particular service, corps or branch.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. The number and date of the applicable specification sheet (see 1.2 and 2.1).

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d. When a first article is not required (see 3.2).

e. Packaging requirements (see 3.12, 5.1 and 6.6).

f. When special marking is required (see 3.10).

6.2.1 Contract data requirements. Data conforming to Department of Defense Data Item Descriptions DI-R-4803, DI-R-4805, DI-T-4901, DI-T-4902, DI-T-4903 and DI-T-4904 will usually be required for delivery in connection with this specification for Marine Corps procurements. For other than Marine Corps procurement, the cognizant procuring activity will provide comparable data item descriptions. When so required, such data will be specified for delivery on a DD-1423 included in the contract.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first production sample or a standard production item from the contractor's current inventory as specified in 4.4. The first article should consist of one completed item. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.8.

6.5 Subject term (key word) listing.

Integral prong
Registering prong
Zinc Aluminum
Nickel
Silver
Nickel Silver
Nitric Acid

6.6 Suggested unit packaging.

6.6.1 Army branch of service insignia. Twenty (20) Army branch of service insignia of one type only will be mounted on a white sulfite board card, 6-1/2 inches wide by 5 inches (16.51 cm by 12.7 cm) and not less than 0.011 inch (0.028 cm) thick. Each card will then be packaged in a snug-fitting transparent polyethylene bag. The polyethylene film will be a minimum of 0.002 inch (0.005 cm) thick. All seams and closures will be heat sealed. Prior to or during the heat sealing operation, excess air within the bag will be expelled.

6.6.2 Intermediate packaging. Two hundred (200) Army branch of service insignia, fifty pair (50) Air Force branch of service insignia, one hundred (100) Female, Army hat insignia, forty (40) Marine Corps service cap insignia, fifty (50) Marine Corps collar branch of service or one hundred (100) Marine Corps garrison cap insignia, of one type only, will be packaged in a setup paperboard box. Box closure will be in accordance with the appendix of the container specification. As an

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alternate method, setup boxes may be securely closed with strips of 1/2 inch (12.7 cm) minimum width pressure-sensitive, filament reinforced tape. A minimum of two strips of pressure-sensitive tape will be used. Each strip will extend not less than 2 inches (5.08 cm) onto each box panel adjacent to the seams of the box.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - IH
Air Force - 45
Navy - NU

Preparing activity:
Army - IH

Review activities:
Air Force - 32, 82, 99
Navy - MC
Coast Guard - CG
DLA - CT

(Project No. 8455-0841)

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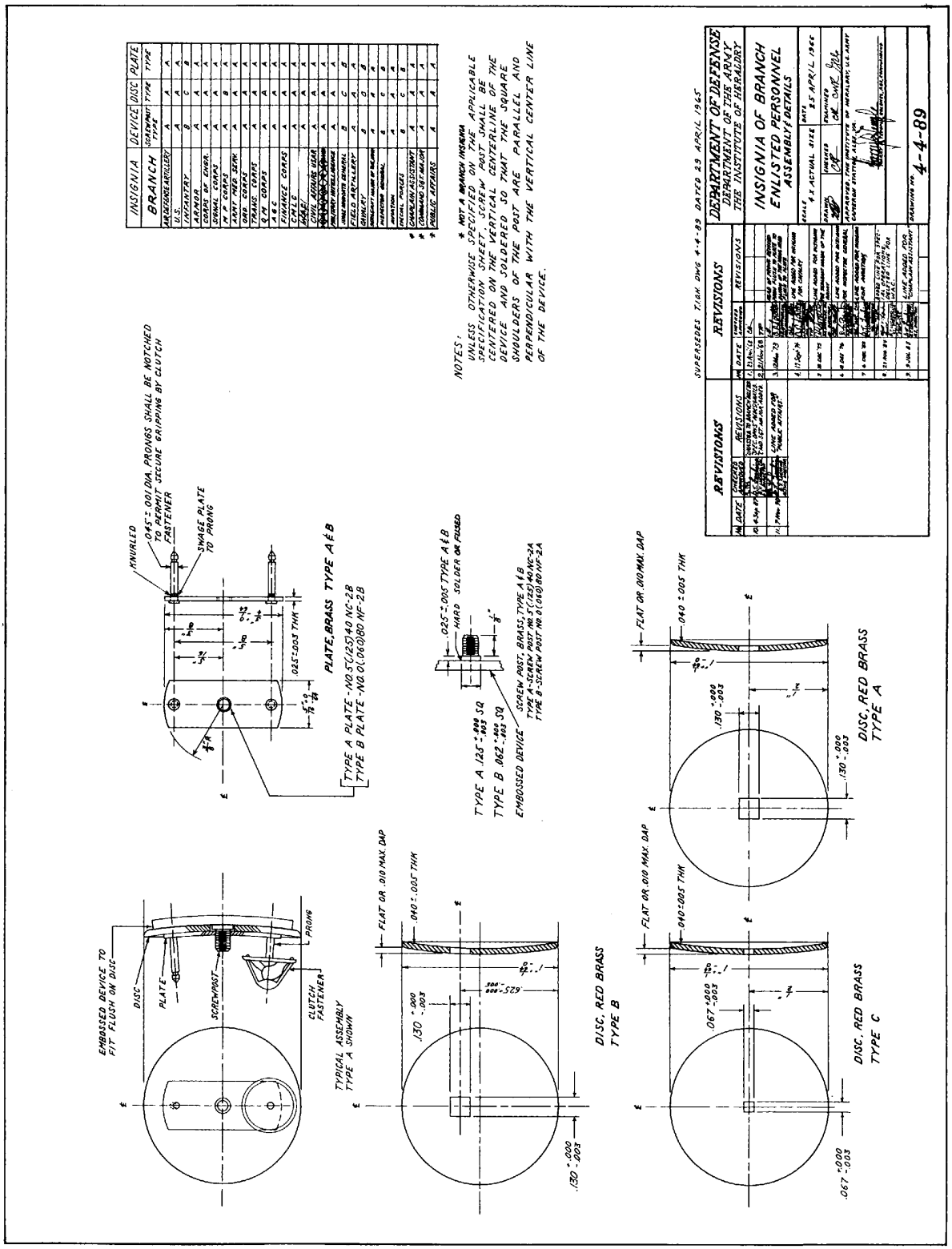


FIGURE 1. Insignia of Branch, Enlisted Personnel, Assembly and Details

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-15665J	**2. DOCUMENT DATE (YYMMDD)** 980406

3. DOCUMENT TITLE **INSIGNIA, BRANCH OF SERVICE, HAT, GARRISON CAP AND SERVICE CAP, GENERAL SPECIFICATION FOR**

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Institute of Heraldry, US Army Aircraft Division	b. TELEPHONE *Include Area Code)* (1) Commercial (703) 806-4990 (2) AUTOVON 656-4990
c. ADDRESS *(Include Zip Code)* Technical and Production Division 9325 Gunston Road, RMS11 Fort Belvoir, VA 22060-5579	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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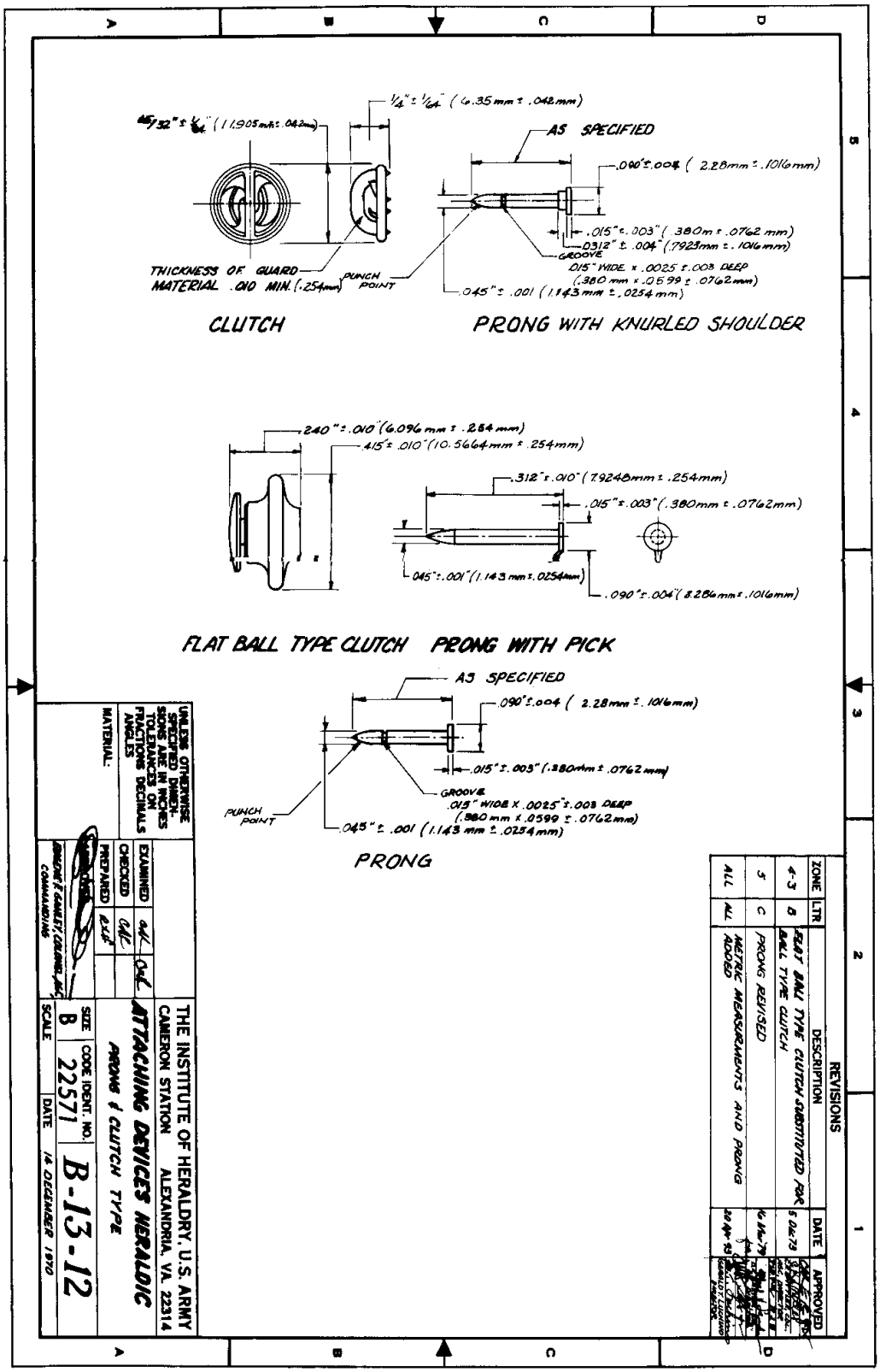


FIGURE 2. Attaching Devices, Heraldic, Prong & Clutch Type

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